

Computer Graphics

Week 1, Day 2

Let's start here



2 activities

- Assuming no prior knowledge of how a contemporary computer graphics pipeline works
- We are going to create our own simple method to **model** and **render** this



Note- this will inspire the methods we explore later in the class, but it won't look exactly the same

Digging deeper

- **Model:** How do we describe the cube?
- **Rendering:** How do we then translate this model into an image?



Step by step- Basic details

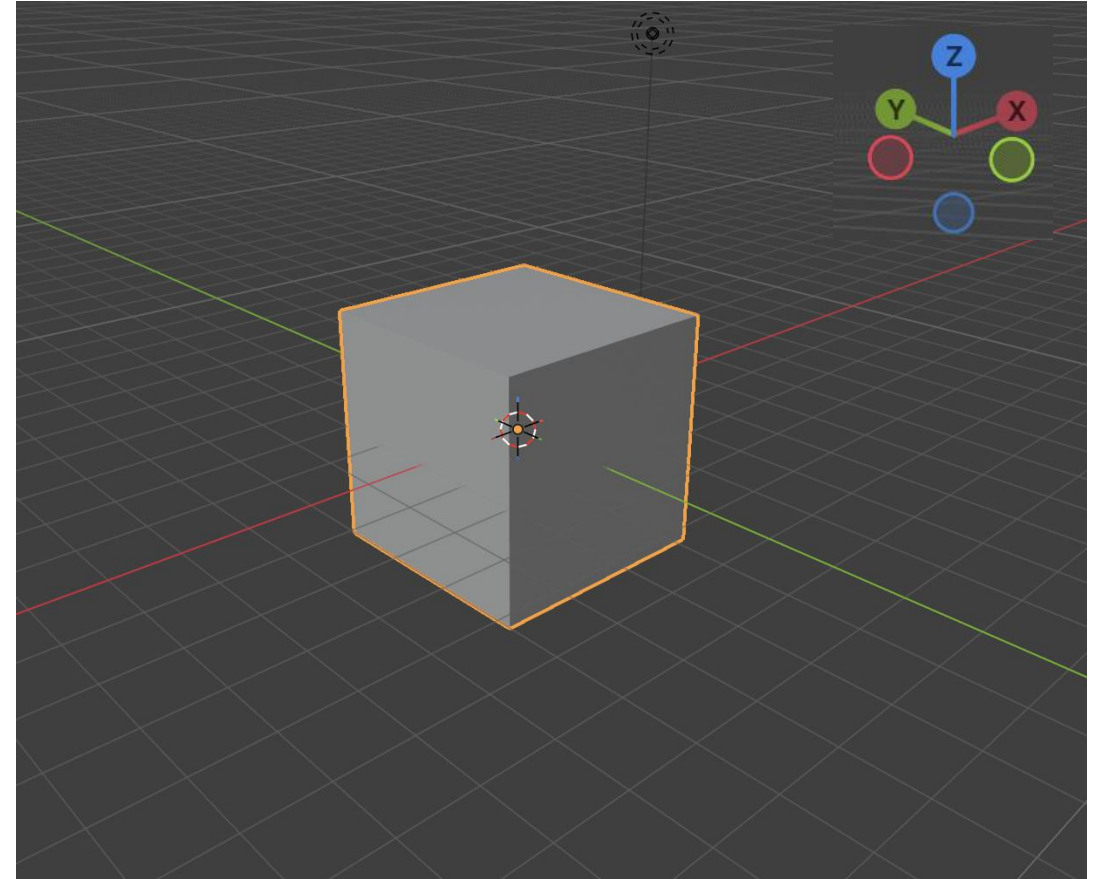
Suppose our cube is:

- Centered at the origin (0, 0, 0)
- Has dimensions 2x2x2
- Edges are aligned with x, y, z axis

`Cube(c_x, c_y, c_z, dim)`

`Cube(0, 0, 0, 2)`

You 'know' enough information about the cube's shape to make it



Step by step- Basic details

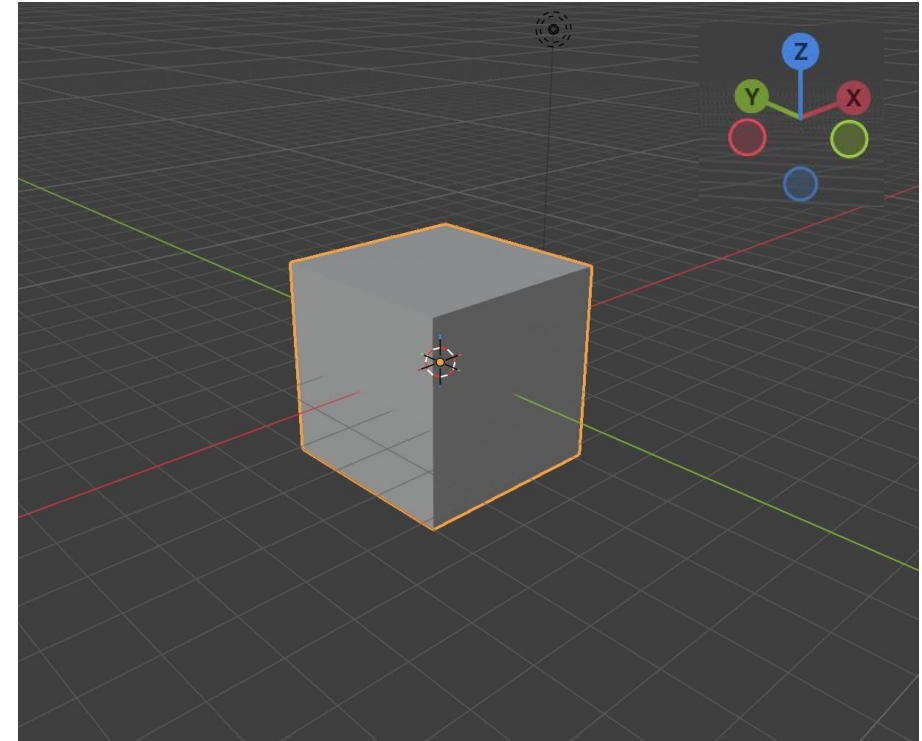
Suppose our cube is:

- Centered at the origin $(0, 0, 0)$
- Has dimensions $2 \times 2 \times 2$
- Edges are aligned with x, y, z axis

This is defines a cube,

but this description does not generalize to other objects (e.g. pyramid)

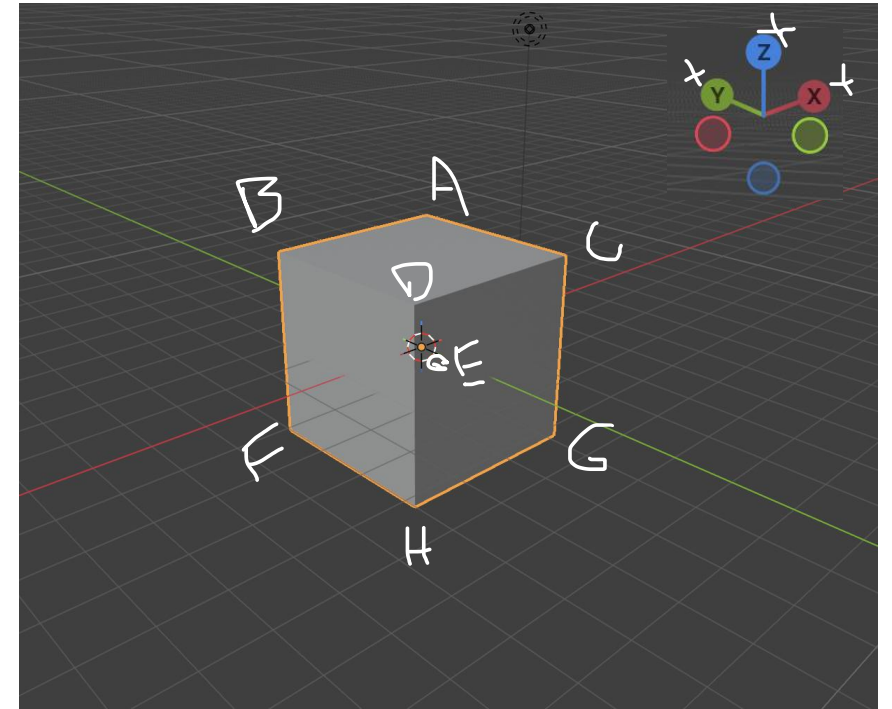
What if we started by defining the coordinates of the corners



Step by step- Basic details

Suppose our cube is:

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- Has dimensions 2x2x2
- Edges are aligned with x, y, z axis

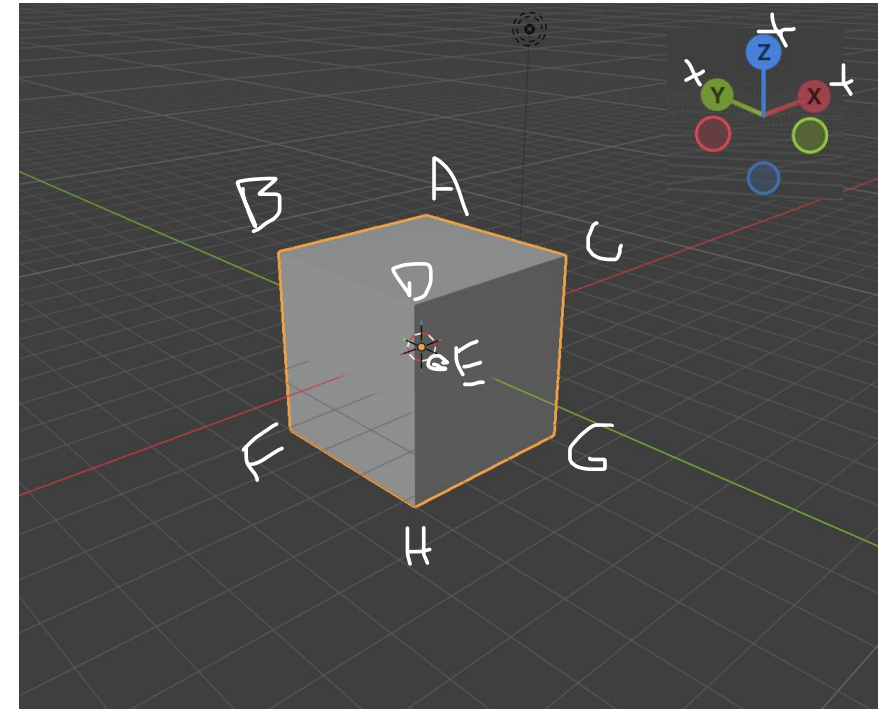


QUESTION: What are the coordinates of the cube vertices?

Step by step- Basic details

Suppose our cube is:

- Centered at the origin (0, 0, 0)
- Has dimensions 2x2x2
- Edges are aligned with x, y, z axis



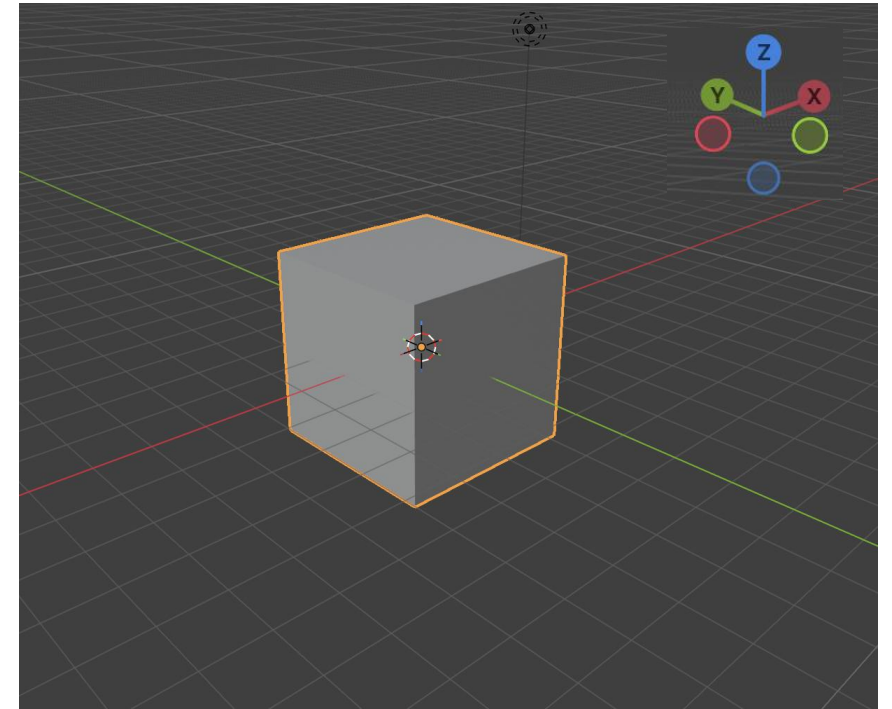
QUESTION: What are the coordinates of the cube vertices?

A: (1, 1, 1)	E: (1, 1, -1)
B: (-1, 1, 1)	F: (-1, 1, -1)
C: (1, -1, 1)	G: (1, -1, -1)
D: (-1, -1, 1)	H: (-1, -1, -1)

Step by step- Basic details

Suppose our cube is:

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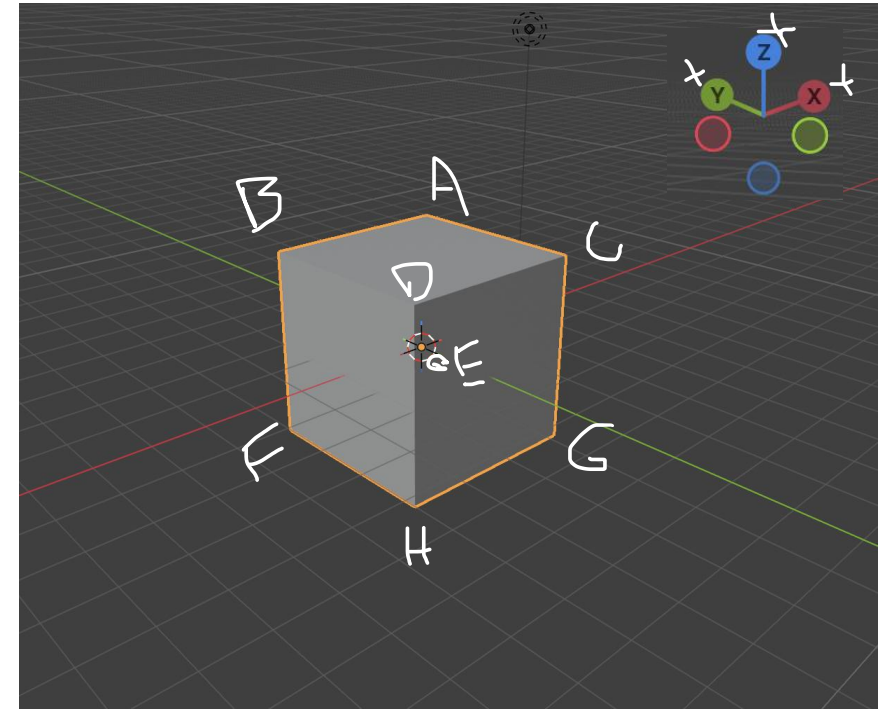
Is this enough to digitally model the cube?

Maybe it would just be a bunch of points in space....

Step by step- Basic details

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QUESTION: What are the coordinates of the cube vertices?

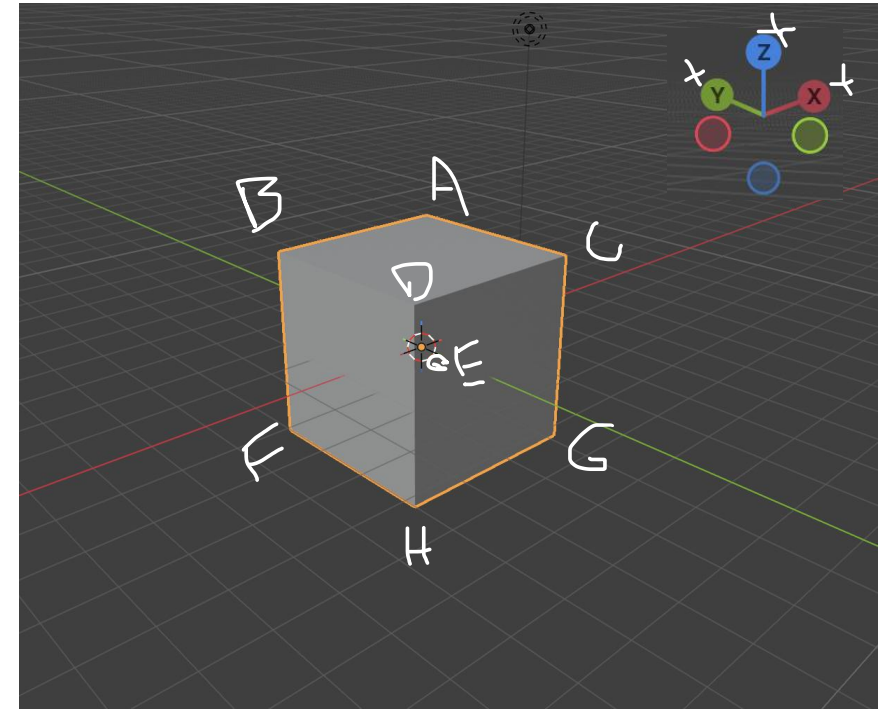
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QUESTION: What are the edges?

Step by step- Basic details

Suppose our cube is:

- Centered at the origin (0, 0, 0)
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QUESTION: What are the coordinates of the cube vertices?

A: (1, 1, 1)	E: (1, 1, -1)
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D: (-1, -1, 1)	H: (-1, -1, -1)

QUESTION: What are the edges?

AB,	CD,	EF,	GH,
AC,	BD,	EG,	FH,
AE,	CG,	BF,	DH

Step by step: from model to rendering

- Now we have a digital description of a cube

VERTICES

A: (1, 1, 1)	E: (1, 1, -1)
B: (-1, 1, 1)	F: (-1, 1, -1)
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D: (-1, -1, 1)	H: (-1, -1, -1)

EDGES

AB, CD, EF, GH,
AC, BD, EG, FH,
AE, CG, BF, DH

- How do we draw this 3D cube on a 2D (flat) image?

Step by step: from model to rendering

- Now we have a digital description of a cube

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A: (1, 1, 1)	E: (1, 1, -1)
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EDGES

AB, CD, EF, GH,
AC, BD, EG, FH,
AE, CG, BF, DH

- How do we draw this 3D cube on a 2D (flat) plane (our image)?
- Basic strategy:
 - Figure out where the 3D points should be placed on 2D image
 - Draw lines to connect the right points together

Step by step: from model to rendering

- Now we have a digital description of a cube

VERTICES

A: (1, 1, 1)	E: (1, 1, -1)
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EDGES

AB, CD, EF, GH,
AC, BD, EG, FH,
AE, CG, BF, DH

- How do we draw this 3D cube on a 2D (flat) plane (our image)?
- Basic strategy:
 - Figure out where the 3D points should be placed on 2D image **HOW?**
 - Draw lines to connect the right points together

Perspective



Objects further away look smaller than objects up close

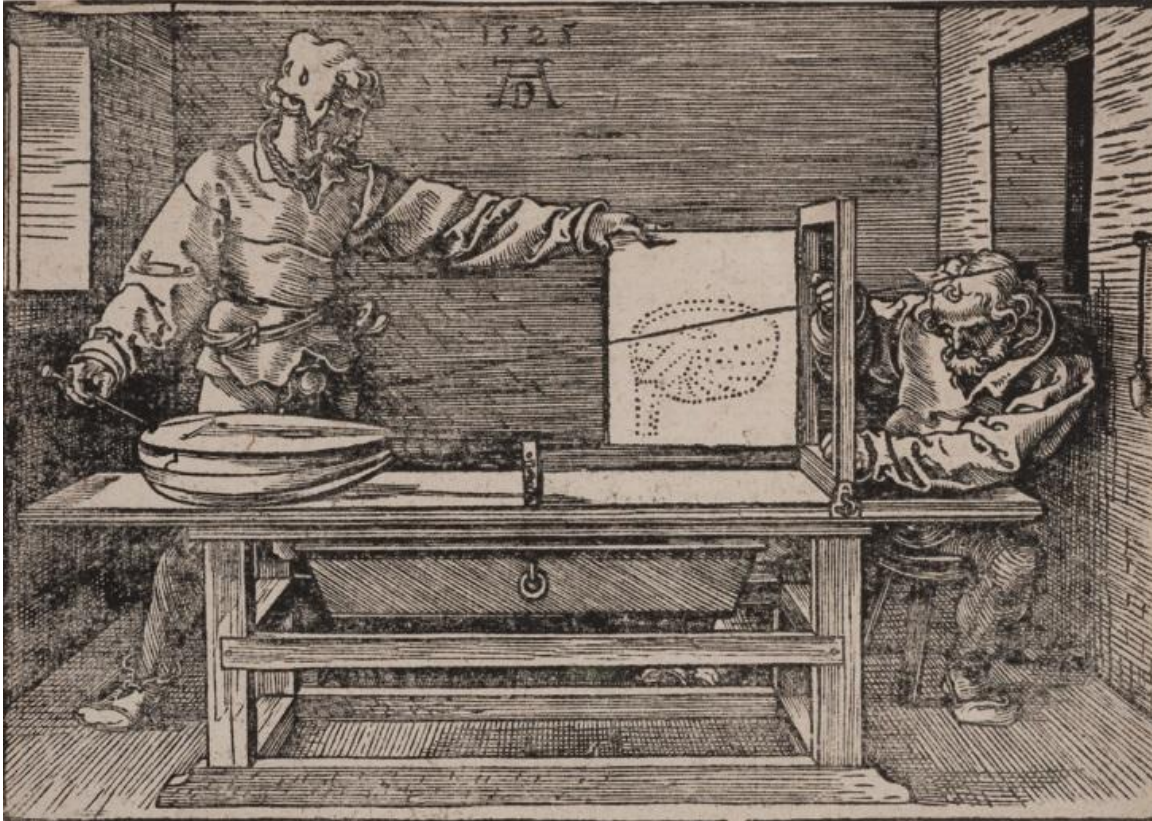
History: Perspective



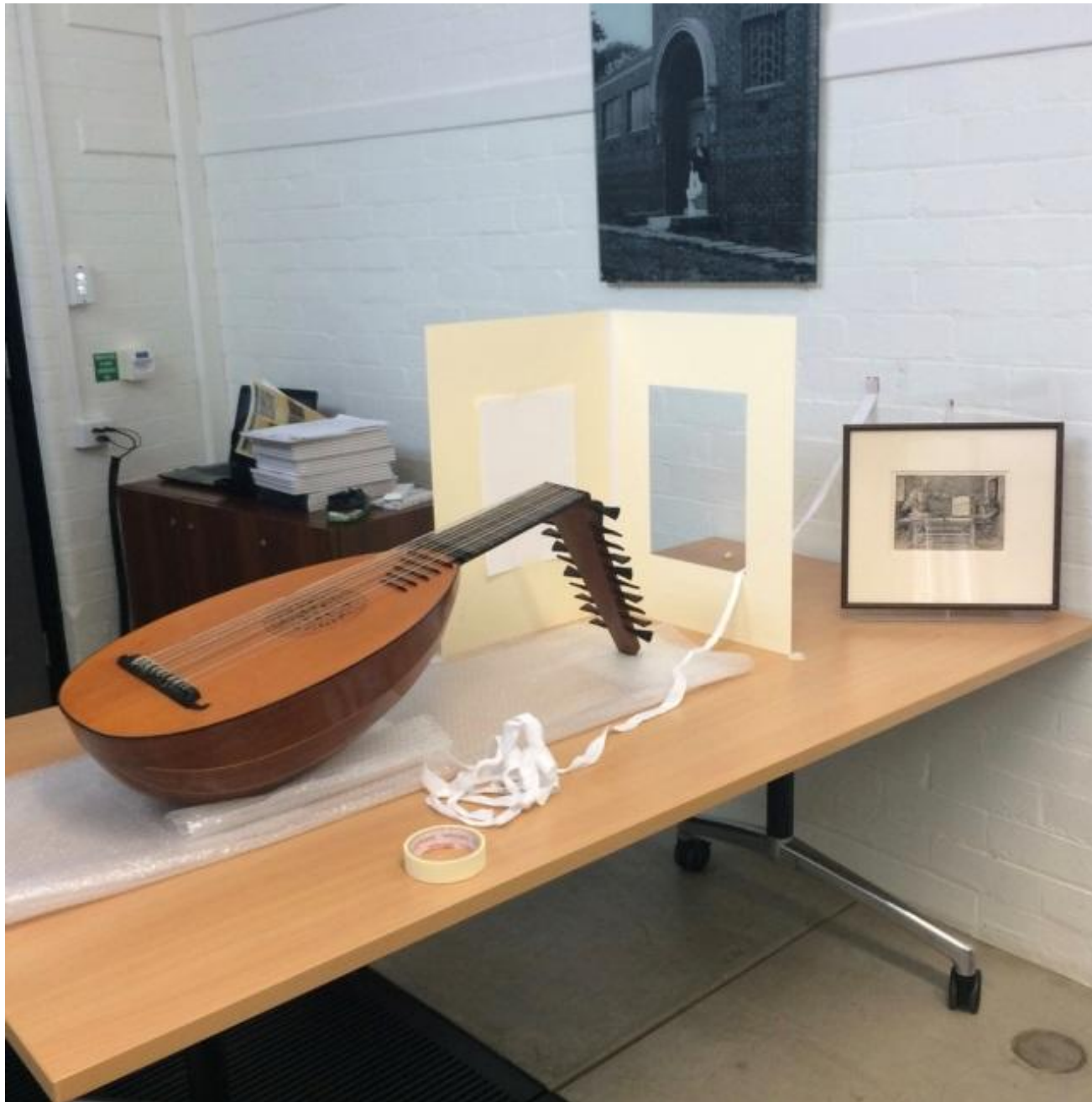


An Artist Using a Perspective Machine, Albrecht Dürer





To paraphrase Dürer's own description, a draftsman uses '*a strong thread*' hammered into the wall to create 'the near point of sight' and places a vertical frame parallel to the wall. Then '*a lute or other object to your liking*' is placed on the opposite end of the table to the wall. The near point of sight is placed on parts of the lute and string attached with hot wax to the frame to mark where the near point of sight passes through the frame. The points that the crossed strings denote are then marked on 'your drawing tablet' creating an accurate dotted outline for the lute.^[2] This complex description visually translates to the seemingly simple diagram shown in the woodcut.



<https://bpb-ap-se2.wpmucdn.com/blogs.unimelb.edu.au/dist/0/63/files/2017/08/pic-4-28n65ps.jpg>

Perspective- pinhole camera

How to Make a Pinhole Film Camera at Home with Some Packaging

🕒 MAY 11, 2020

👤 DALE WILLETT



<https://petapixel.com/2020/05/11/how-to-make-a-pinhole-film-camera-at-home-with-some-packaging/>

Perspective- pinhole camera

https://www.youtube.com/shorts/pFaAM_UElb4

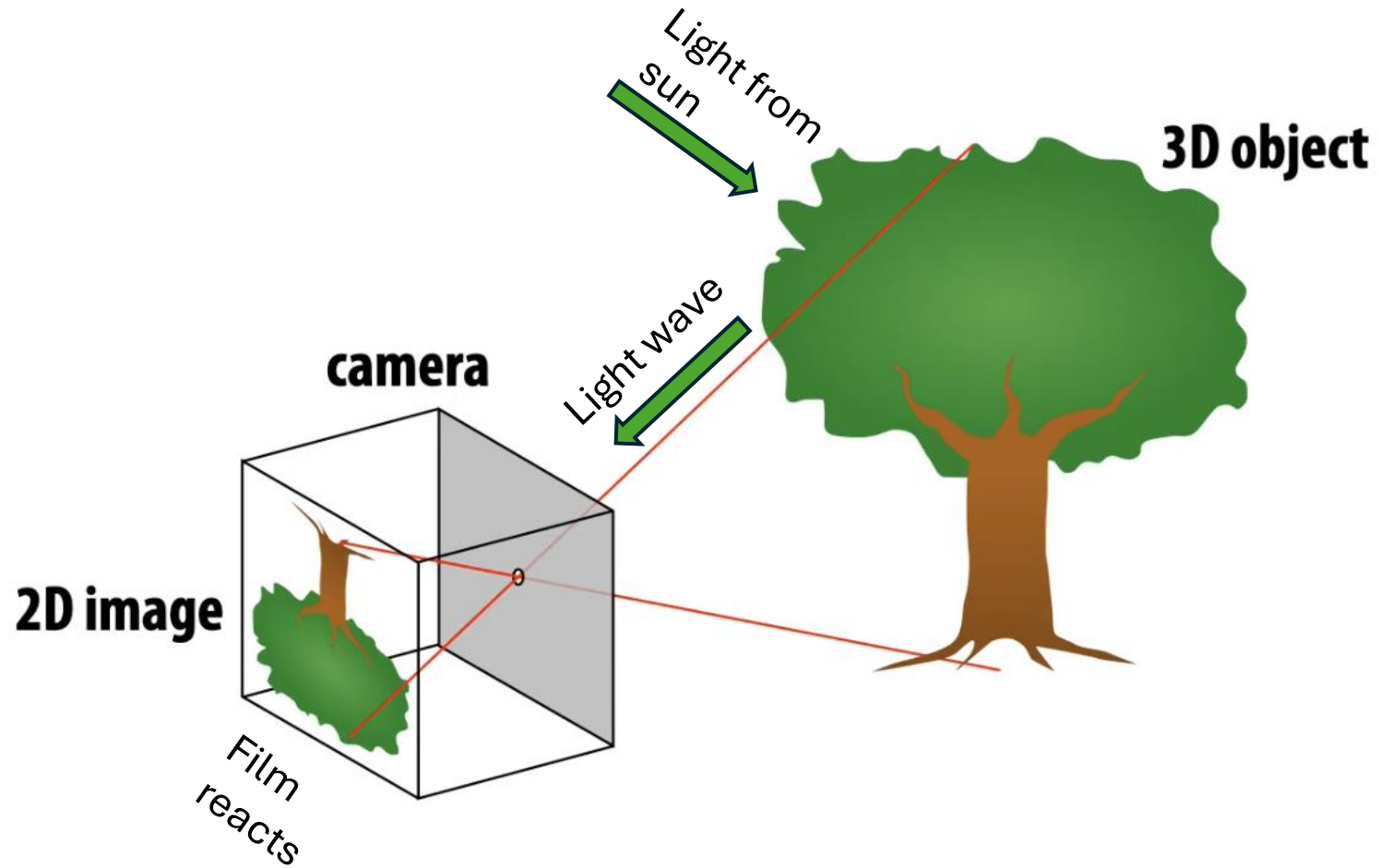


Fun pinhole camera example- just light, no film

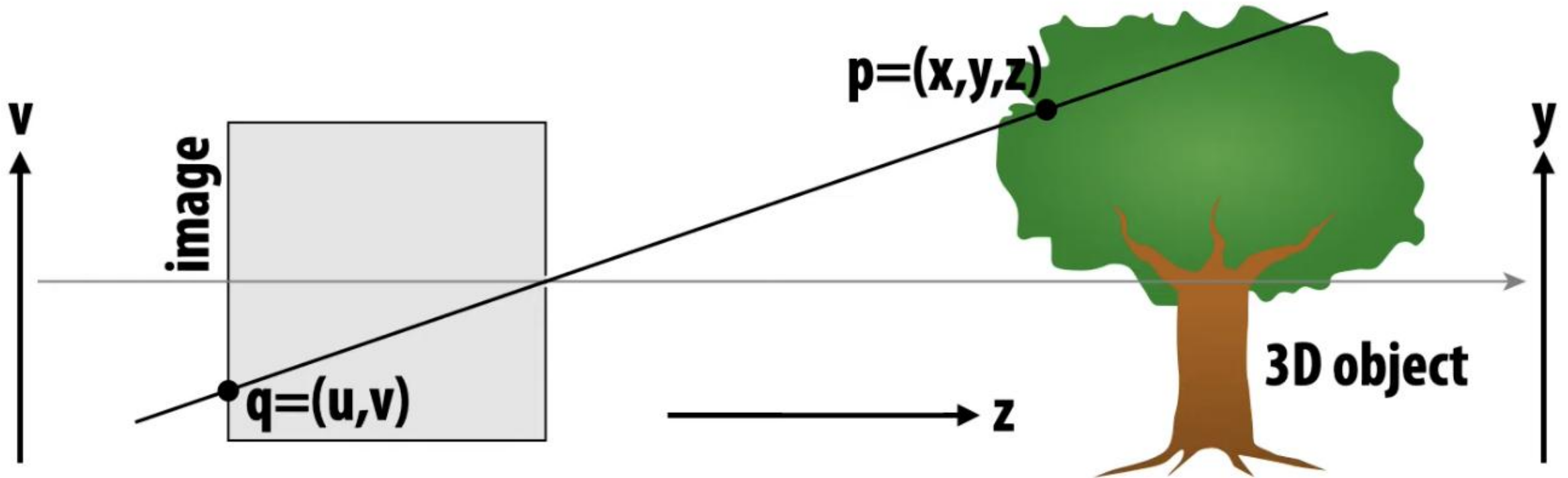


<https://photos.google.com/search/Cghjb2xpbmRlcilKCghjb2xpbmRlciis%2FLDYgzQ4Aw%3D%3D/photo/AF1QipMBo-tgR6dcPyES1IknIMuxfLQS8rRsgP-Pr9U>

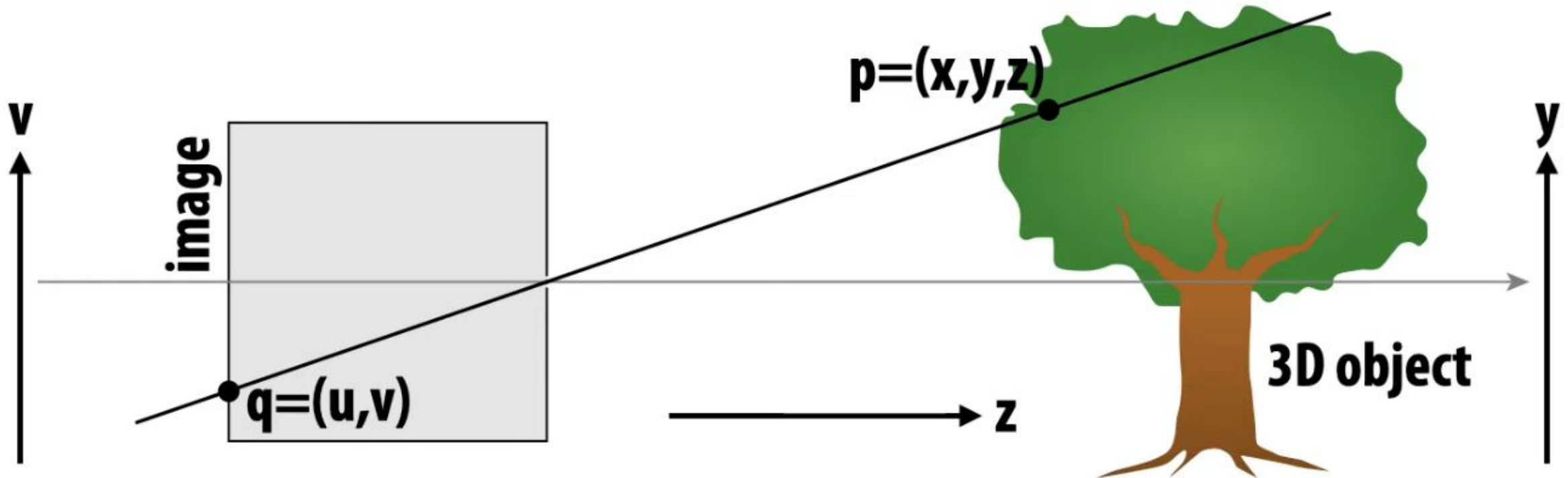
Perspective- pinhole camera



Perspective- pinhole camera, side view

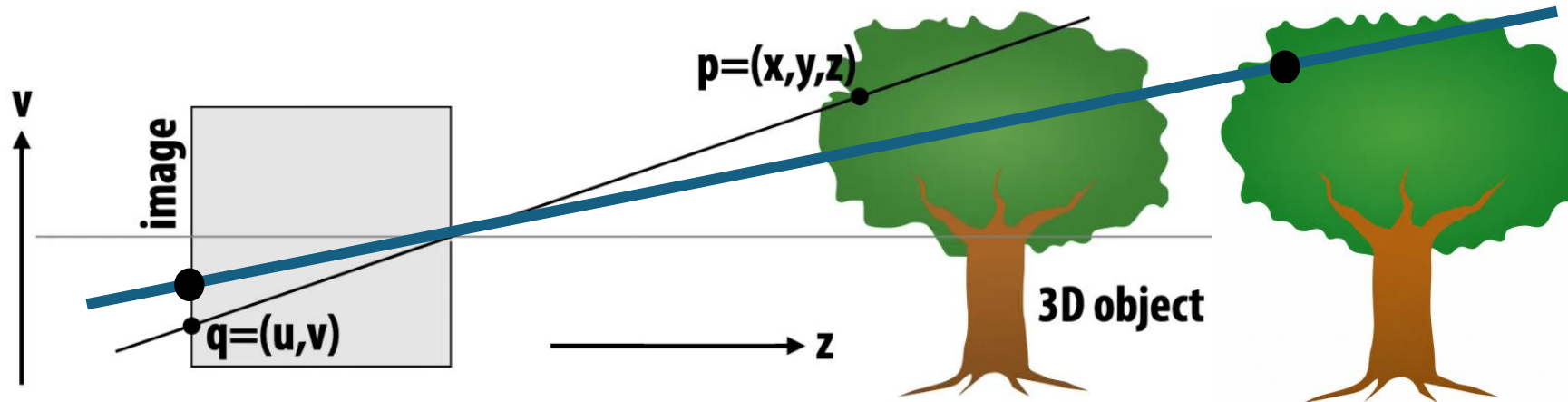


Perspective- pinhole camera, side view



What happens if I move the tree further away?
Does a larger or smaller percent of the back plane of
the pinhole camera get covered by the tree?

Perspective- pinhole camera, side view

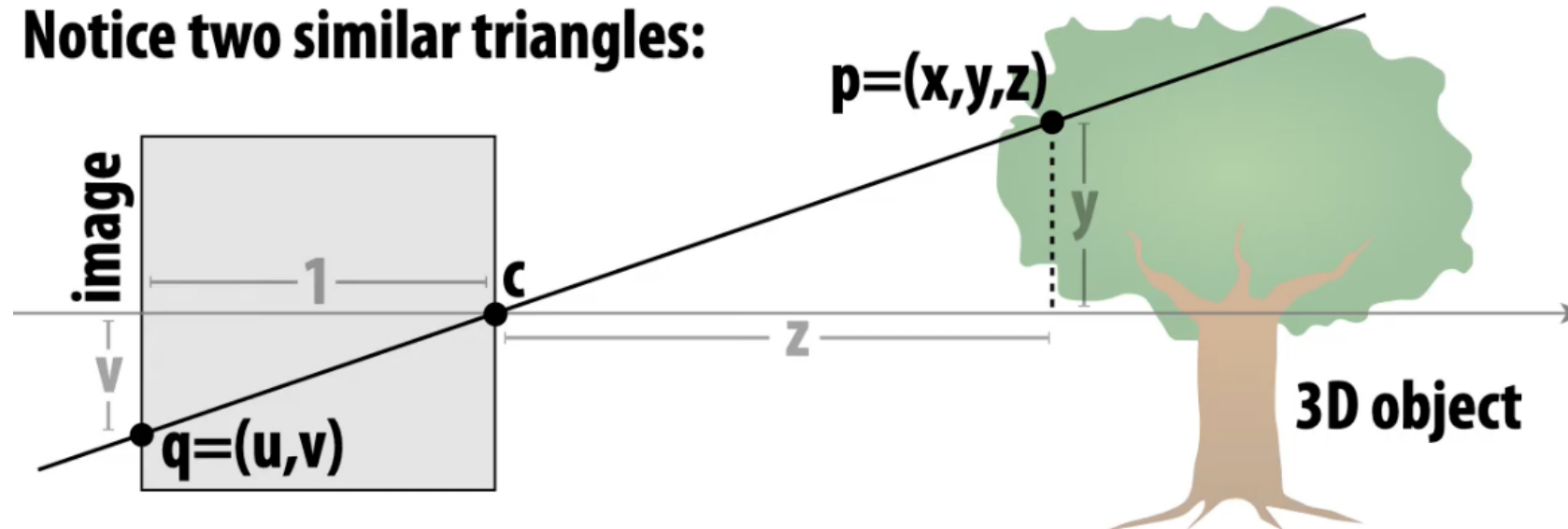


Perspective- pinhole camera, side view

Can we define the relationship between point p at coordinate x,y,z in 3D space
and point q which is at coordinate (u,v) in the 2D image?

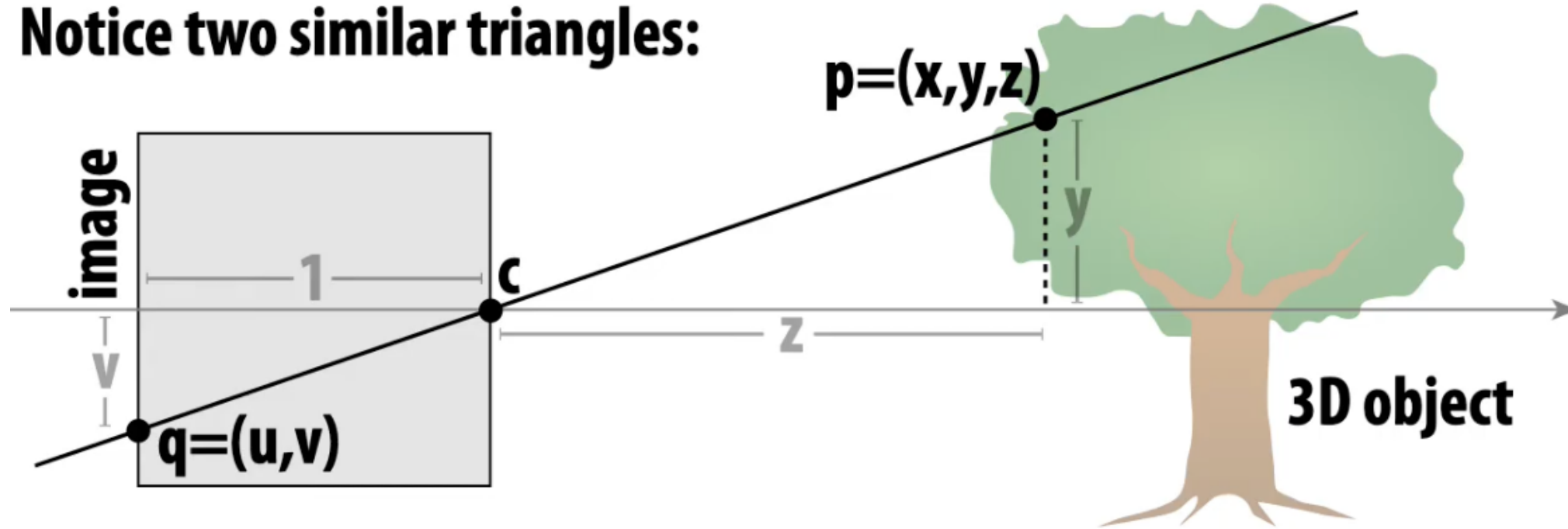
Specifically: what is the relationship between the 'v' coordinate and the 'z' and 'y' coordinate?

Notice two similar triangles:



Perspective- pinhole camera, side view

Notice two similar triangles:

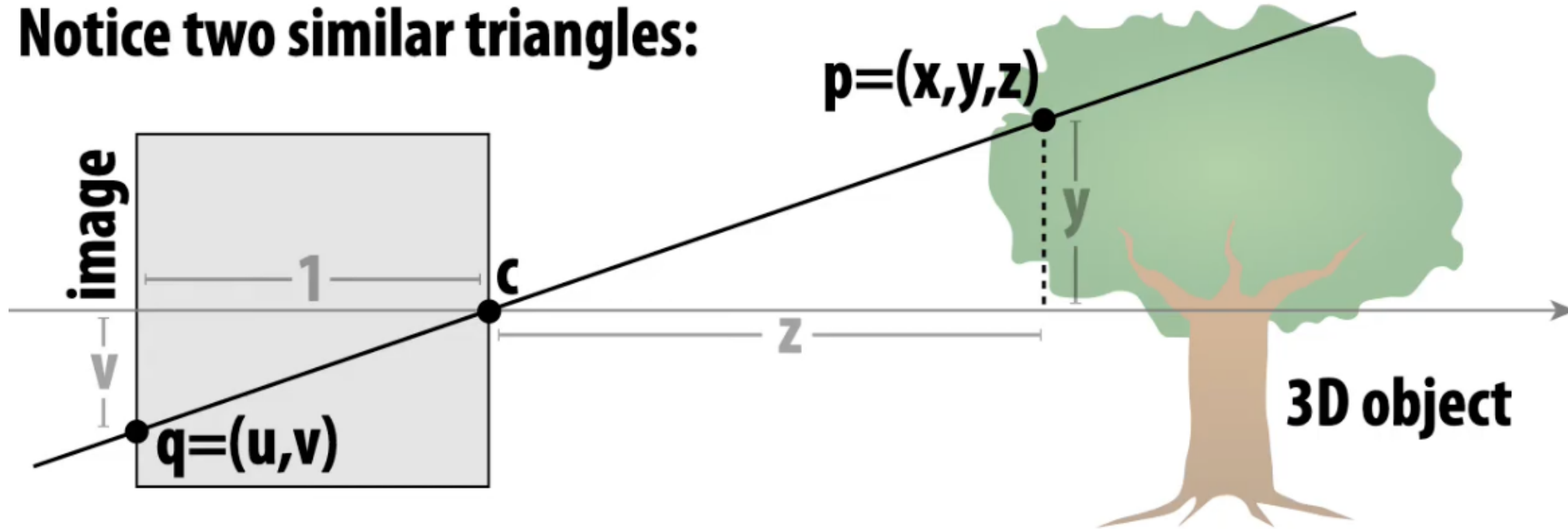


Assume camera is 'unit size' (length 1).
And the origin is at the pinhole

$$v/1 = y/z$$

Perspective- pinhole camera, side view

Notice two similar triangles:



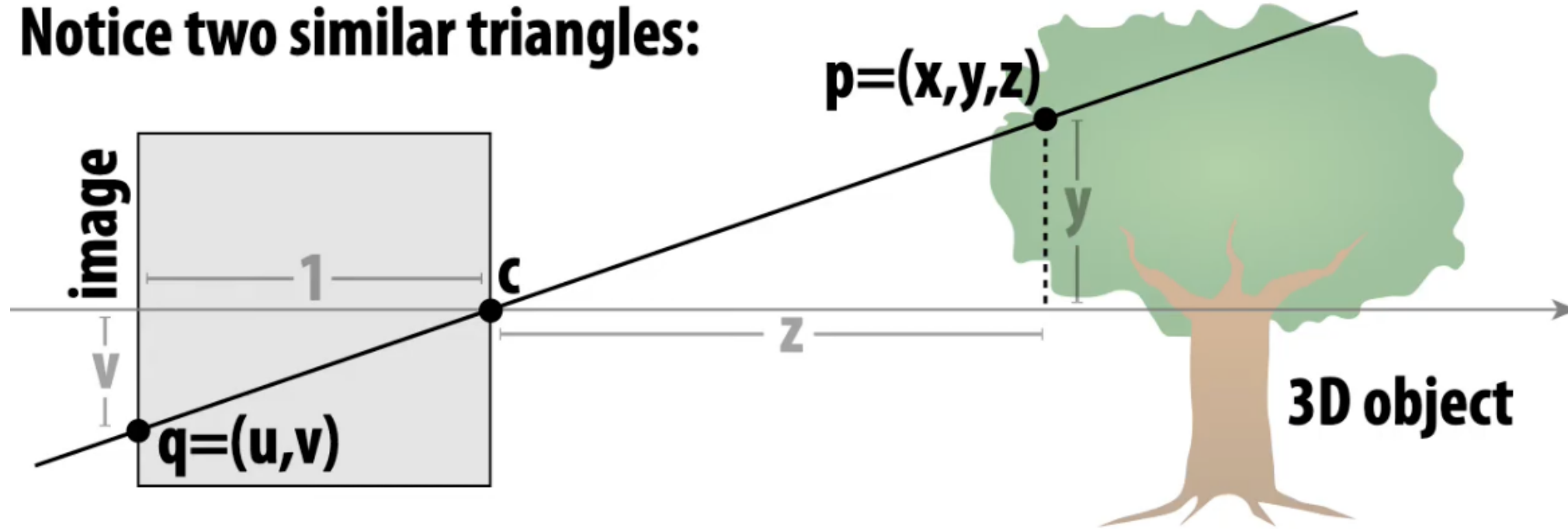
Assume camera is 'unit size' (length 1).
And the origin is at the pinhole

$$v = y / z$$

What is the horizontal coordinate?

Perspective- pinhole camera, side view

Notice two similar triangles:



Assume camera is 'unit size' (length 1).
And the origin is at the pinhole

What is the horizontal coordinate?

$$v = y / z$$

$$u = x / z$$

Recall:

Step by step: from model to rendering

- Now we have a digital description of a cube

VERTICES

A: (1, 1, 1)	E: (1, 1, -1)
B: (-1, 1, 1)	F: (-1, 1, -1)
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D: (-1, -1, 1)	H: (-1, -1, -1)

EDGES

AB, CD, EF, GH,
AC, BD, EG, FH,
AE, CG, BF, DH

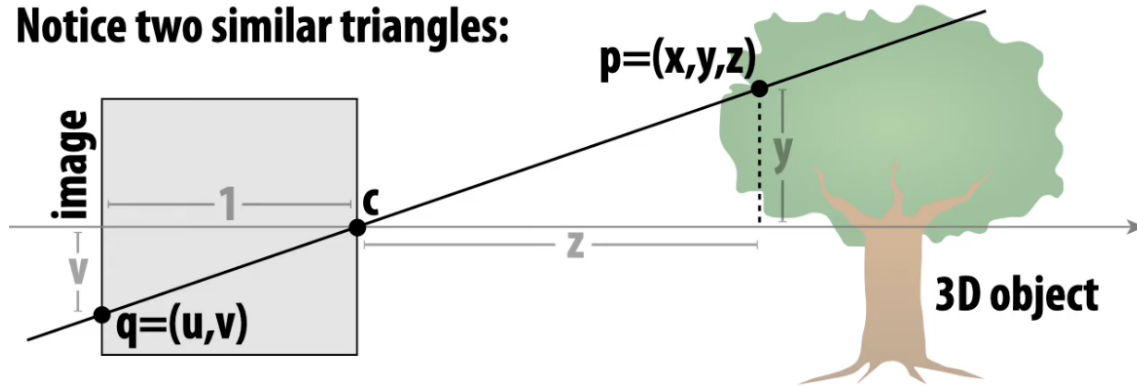
- How do we draw this 3D cube on a 2D (flat) plane (our image)?
- Basic strategy:
 - Figure out where the 3D points should be placed on 2D image
 - Draw lines to connect the right points together

HOW?

Divide x and y by z?

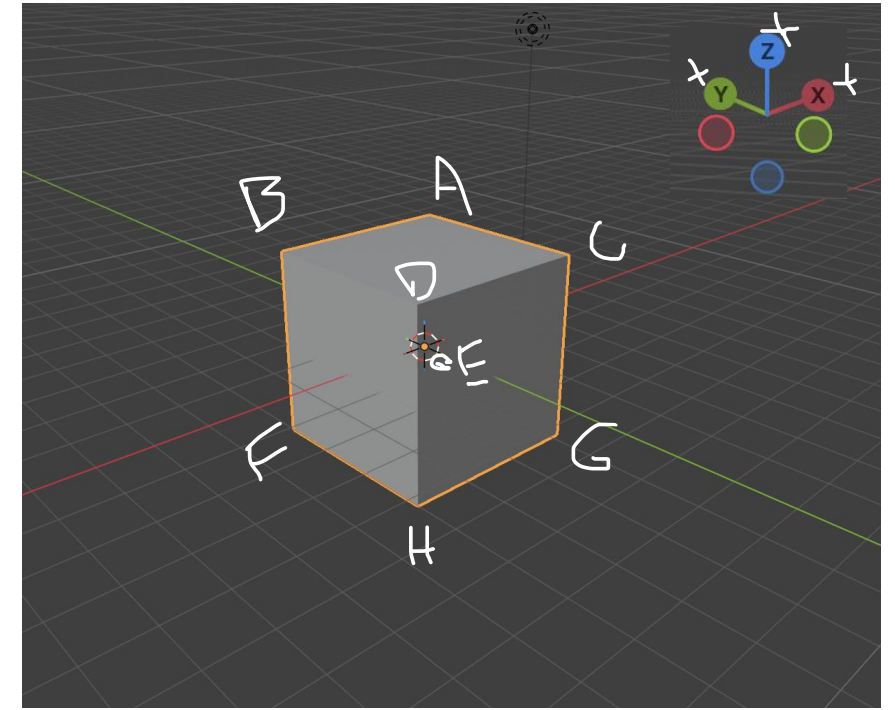
Rendering algorithm part 1

Notice two similar triangles:



$$v = y / z$$

$$u = x / z$$



Cube is centered at origin

We need to know where the camera is

Rendering algorithm part 1

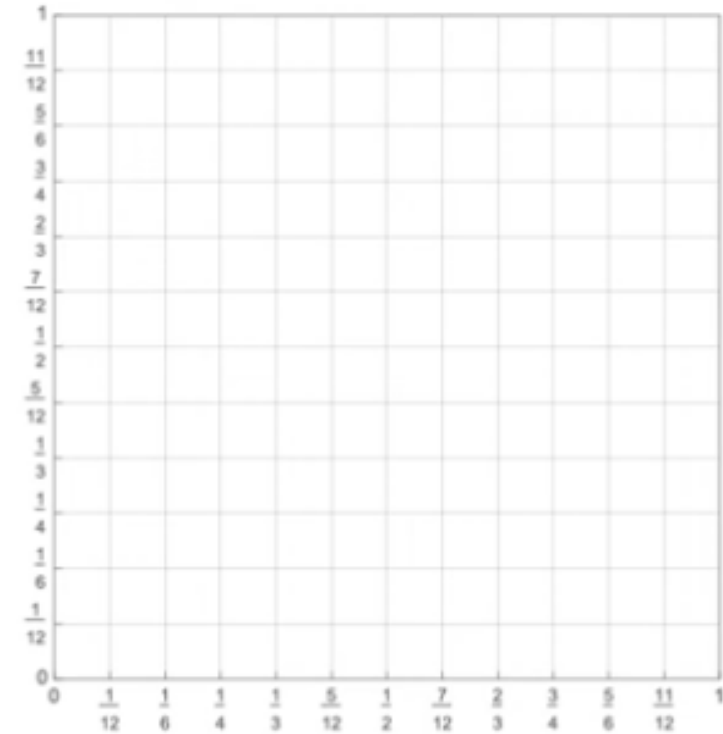
- Assume camera is at $c = (2, 3, 5)$
 - Position it a bit away from the cube so we can see it
- Could we just compute $u = x/z$ and $v = y/z$?

VERTICES

A: (1, 1, 1)	E: (1, 1, -1)
B: (-1, 1, 1)	F: (-1, 1, -1)
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EDGES

AB, CD, EF, GH,
AC, BD, EG, FH,
AE, CG, BF, DH



Rendering algorithm part 1

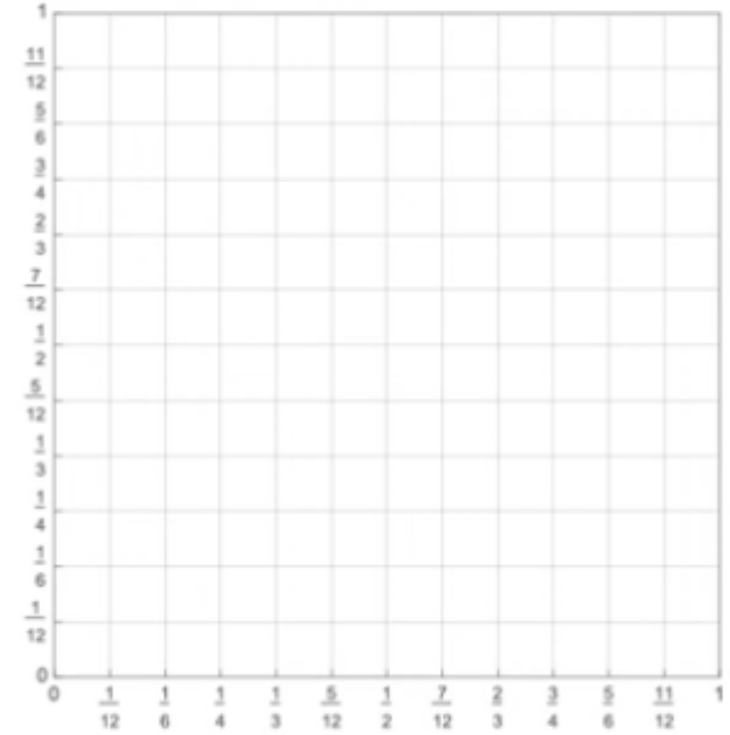
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- We need to convert the coordinates below into coordinates relative to the camera first

VERTICES

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EDGES

AB, CD, EF, GH,
AC, BD, EG, FH,
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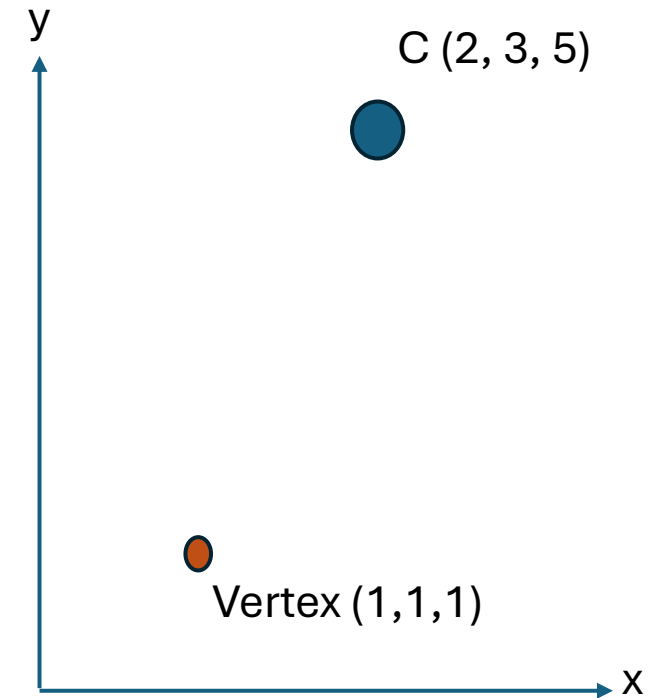


Rendering algorithm part 1

- Assume camera is at $c = (2, 3, 5)$
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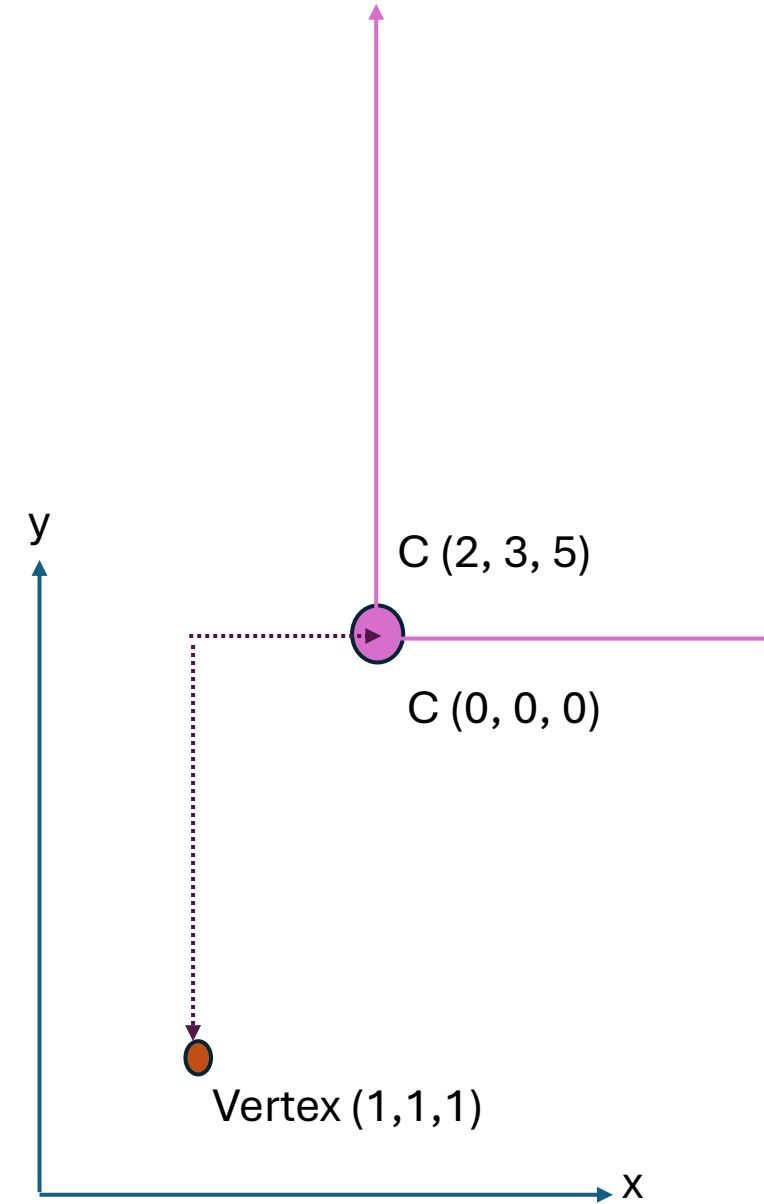
Rendering algorithm part 1

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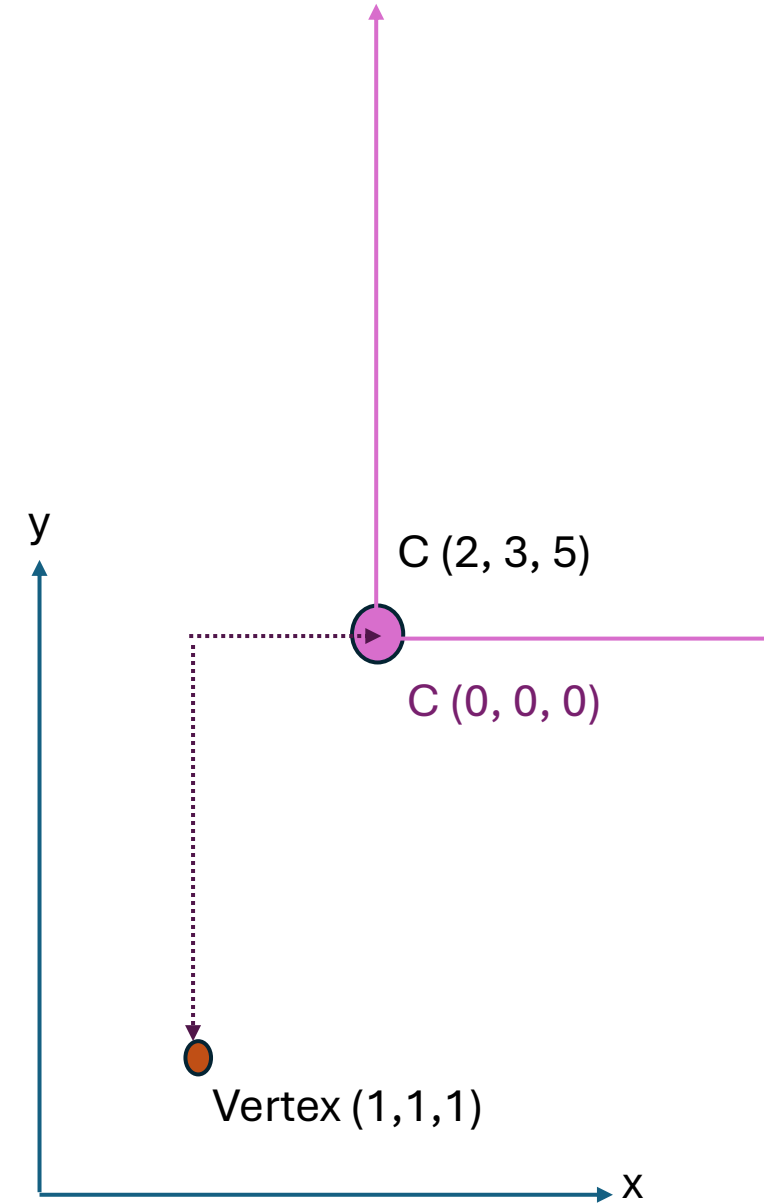
Rendering algorithm part 1

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- What coordinates would the vertex have in the coordinate system based on the camera (purple)



Rendering algorithm part 1

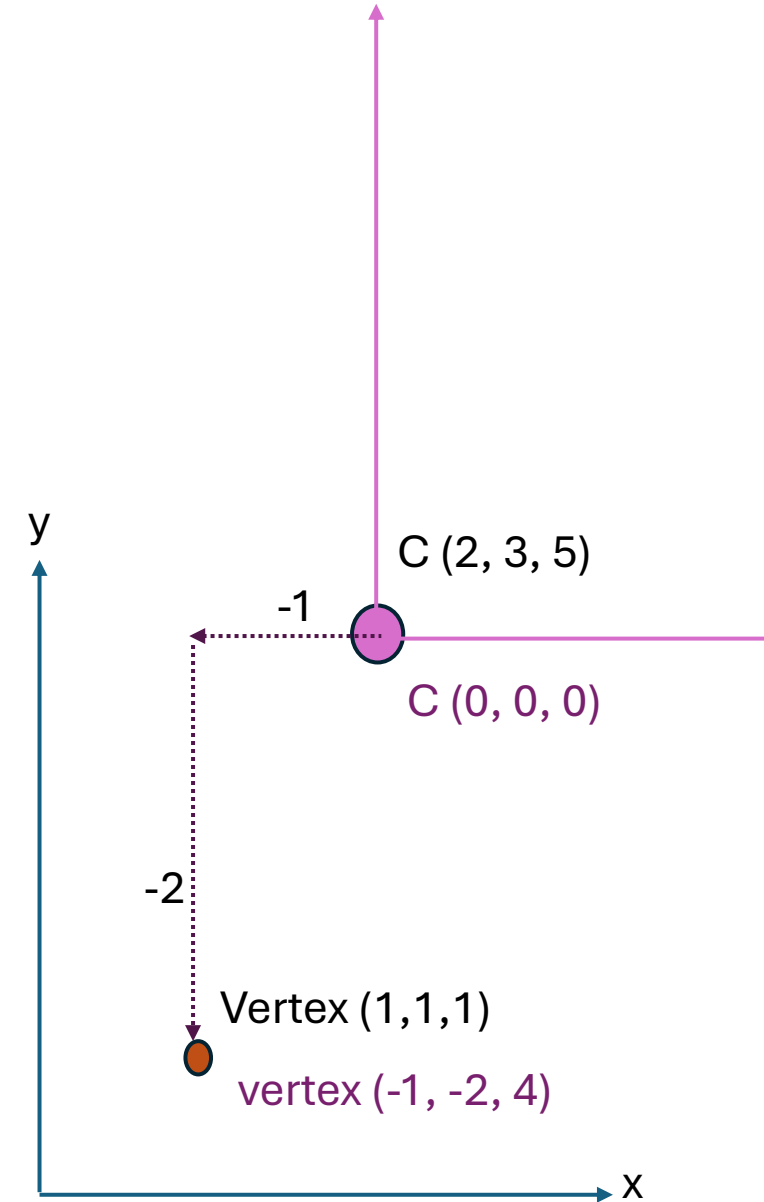
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Rendering algorithm part 1

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- Could we just compute $u = x/z$ and $v = y/z$?
- We need to convert the coordinates below into coordinates relative to the camera first
- What coordinates would the vertex have in the coordinate system based on the camera (purple)

Vertex coordinate – camera coordinate



Rendering algorithm

$C = (2, 3, 5)$

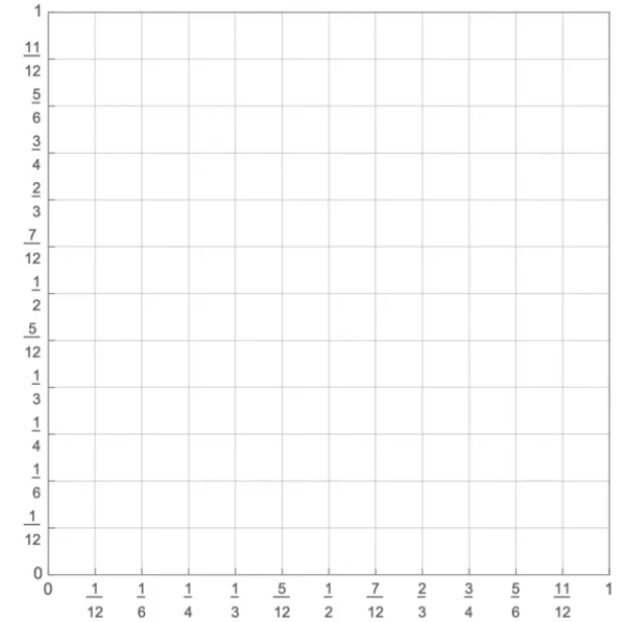
- Repeat for each edge
 - For each vertex in the edge
 - Subtract camera c from vertex (X,Y,Z) to get (x,y,z)
 - Divide x by z to get u
 - Divide y by z to get v
 - Draw line between each vertex $(u1, v1)$ and $(u2, v2)$

VERTICES

A: (1, 1, 1)	E: (1, 1, -1)
B: (-1, 1, 1)	F: (-1, 1, -1)
C: (1, -1, 1)	G: (1, -1, -1)
D: (-1, -1, 1)	H: (-1, -1, -1)

EDGES

AB, CD, EF, GH,
AC, BD, EG, FH,
AE, CG, BF, DH



A = ?

B = ?

C = ?

D = ?

E = ?

F = ?

G = ?

H = ?

Divide the class
Compute
coordinates for me
Tell me what to draw

A = ?

B = ?

C = ?

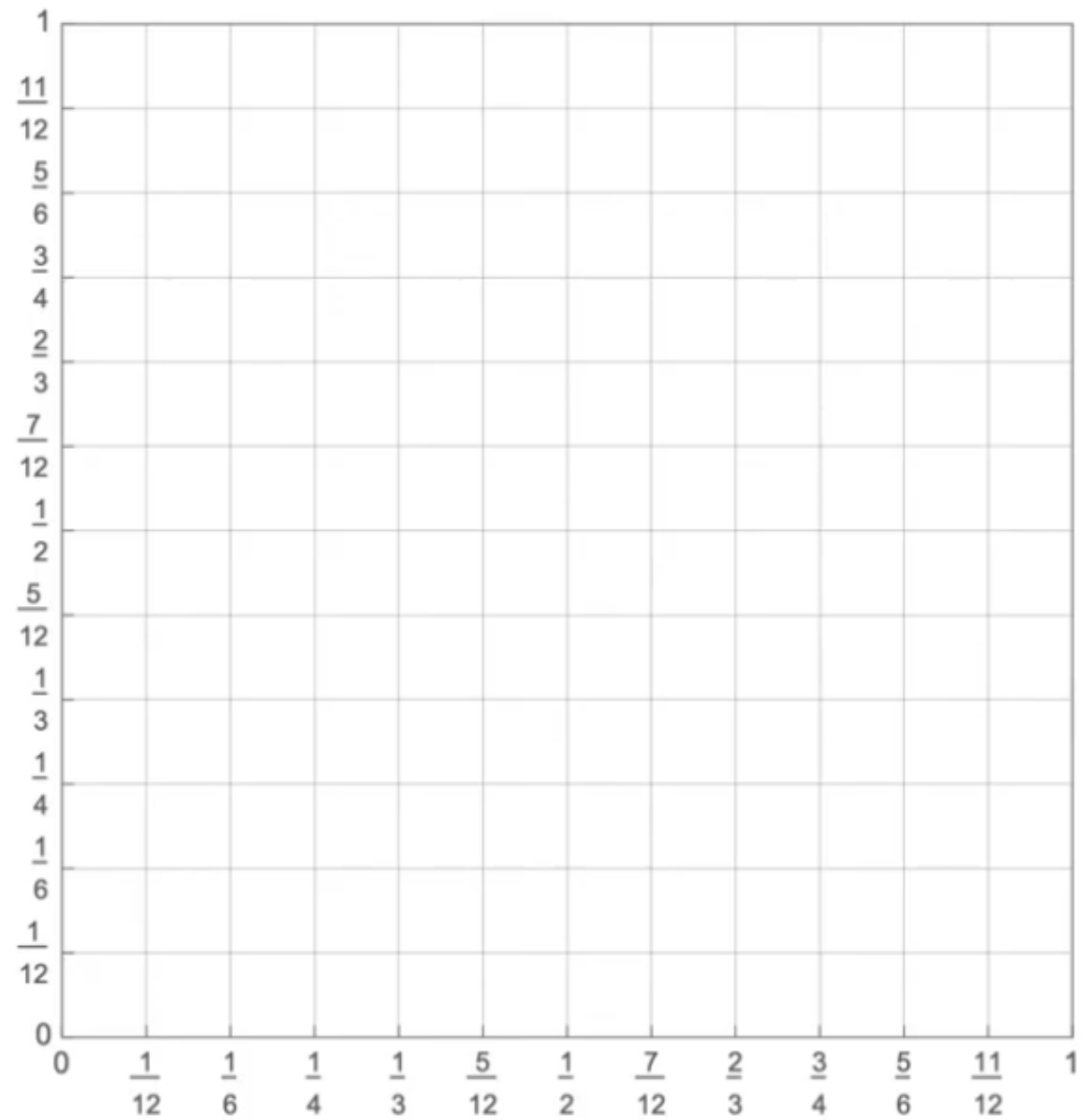
D = ?

E = ?

F = ?

G = ?

H = ?



2D coordinates:

A: $1/4, 1/2$

B: $3/4, 1/2$

C: $1/4, 1$

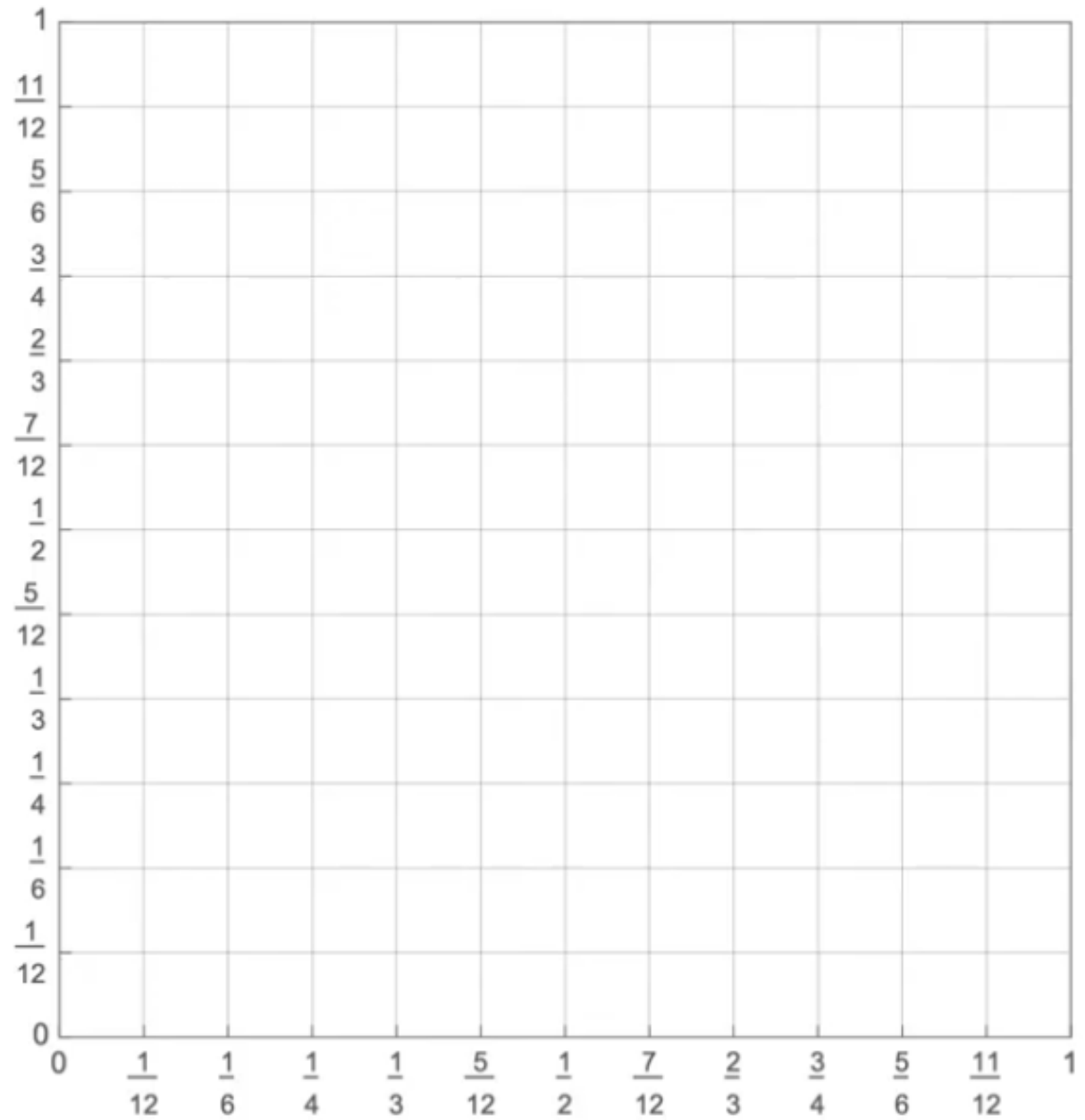
D: $3/4, 1$

E: $1/6, 1/3$

F: $1/2, 1/3$

G: $1/6, 2/3$

H: $1/2, 2/3$



2D coordinates:

A: $1/4, 1/2$

B: $3/4, 1/2$

C: $1/4, 1$

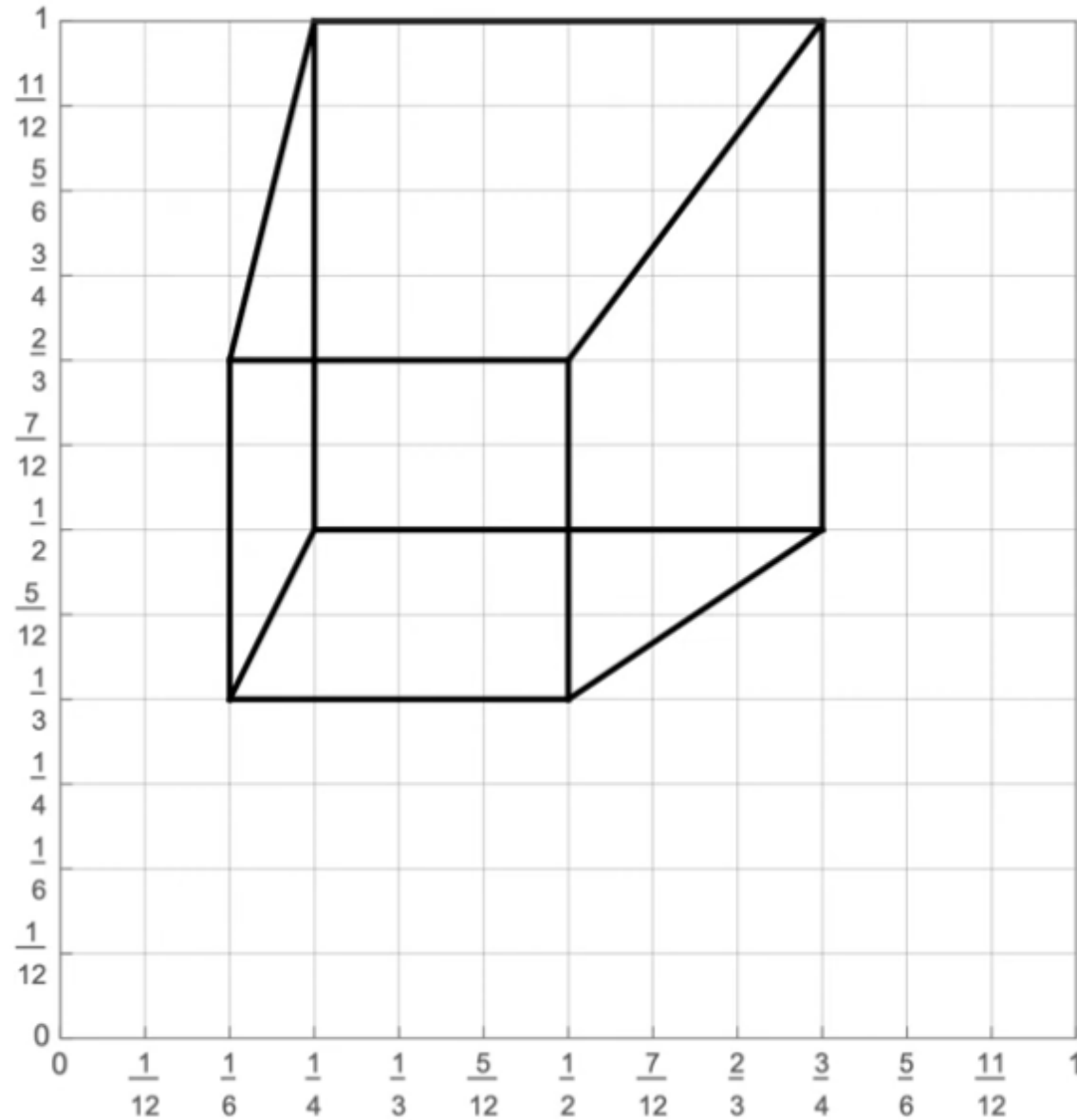
D: $3/4, 1$

E: $1/6, 1/3$

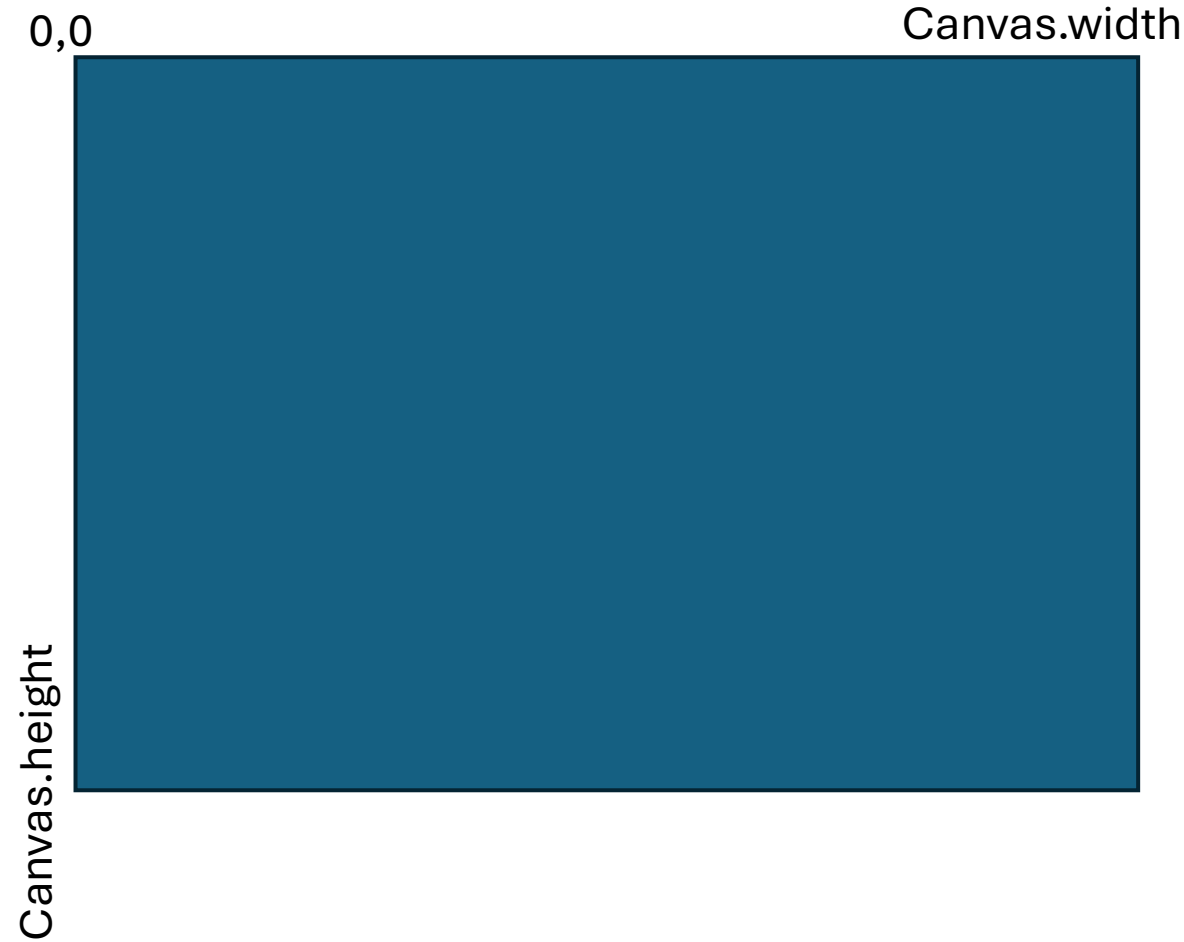
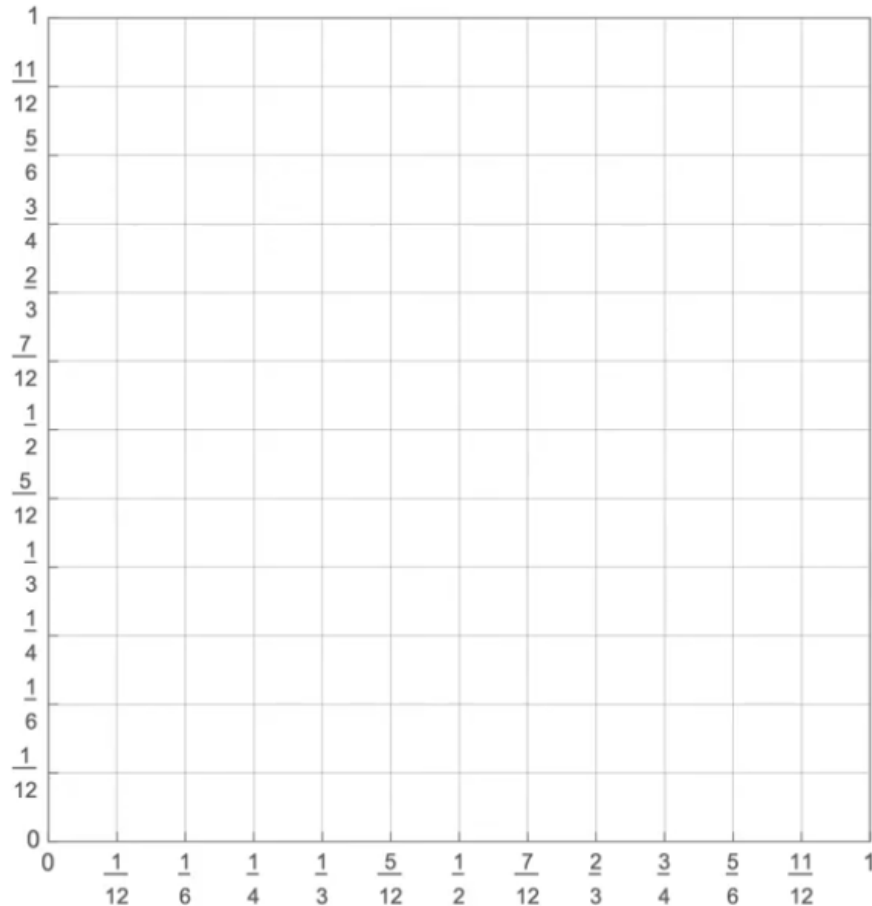
F: $1/2, 1/3$

G: $1/6, 2/3$

H: $1/2, 2/3$



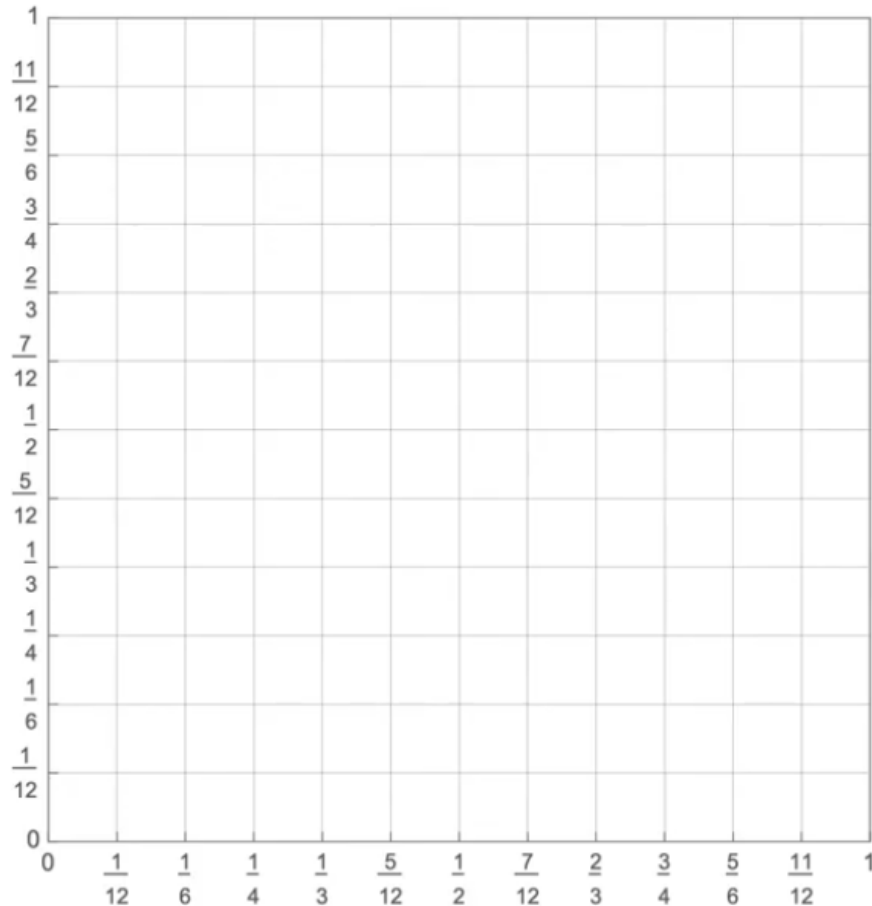
One missing thing



If I got coordinate $(.5, .5)$

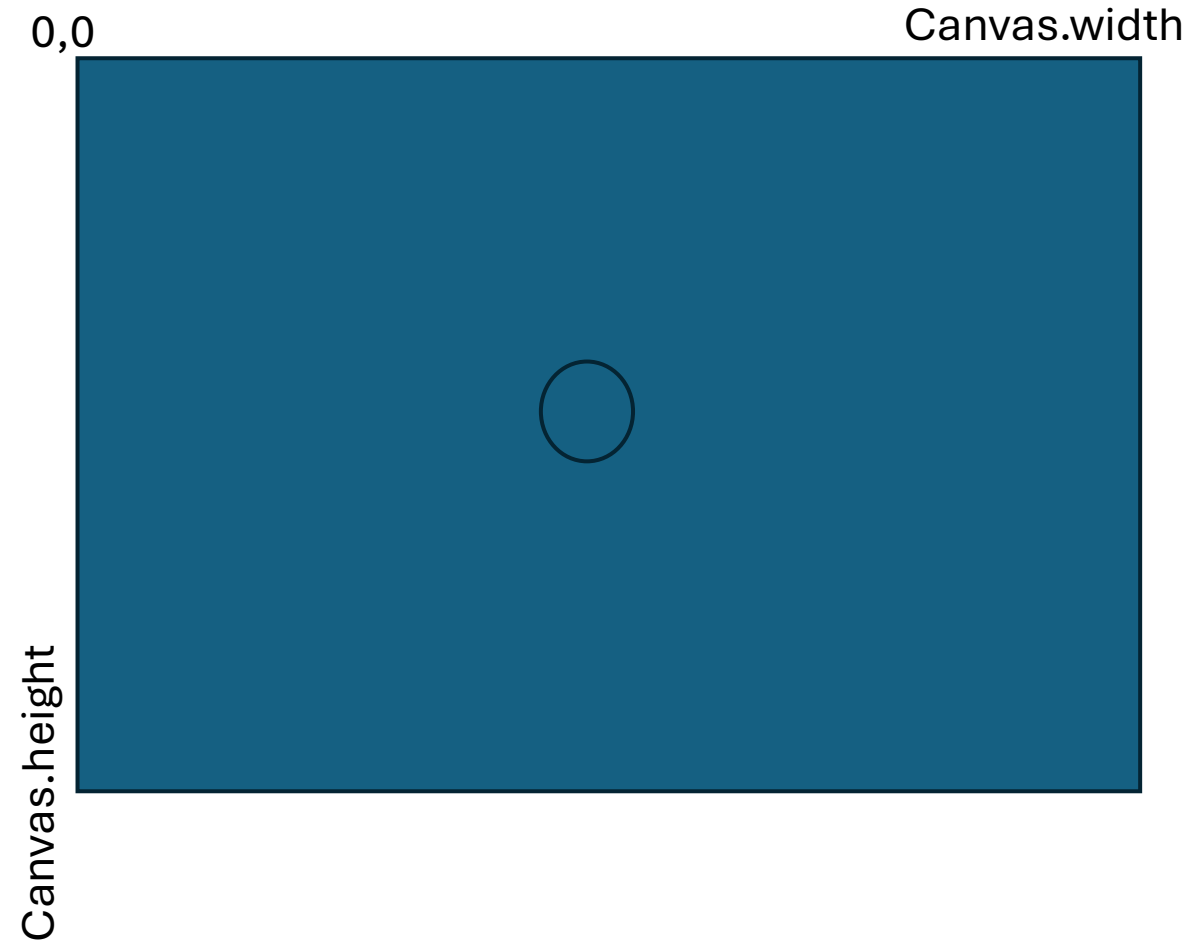
How do I put it on this specific canvas?

One missing thing



If I got coordinate (.5, .5)
How do I put it on this specific canvas?

$\text{Pixel_u} = 5 * \text{Canvas.width}$
 $\text{Pixel_v} = \text{Canvas.height} - .5 * \text{Canvas.height}$



What did we learn here?

- Given a model for a cube
- We defined an algorithm to translatge this model into visual information
- Success?

Let's take a look at some simple code

- ...